William Mallard, PhD

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I bring rigorous biological training and deep computational expertise. I design and build scalable data analysis pipelines that are tightly intertwined with the underlying biology.

Skills

Languages	Python, C, C++, Bash, CUDA, Assembly, Java, MATLAB, R, Unix text processing
Libraries	NumPy, SciPy, Pandas, Matplotlib, Seaborn, PyTorch, Scikit-learn, Scikit-image
Bio Tools	AlphaFold, PyMOL, nmrglue, bedtools, HMMER, Napari, Fiji, BioIO
Systems	Linux/Unix, HPC (SLURM, LSF), PostgreSQL, GPUs, FPGAs, Git
Domains	Computational Biology, Multi-omics, Quantitative Image Analysis, Structural Biology
Methods	Machine Learning, Computer Vision, Agentic RAG (MCP), Real-time Signal Processing

Experience

Harvard University — *Cambridge, MA*
Doctoral Researcher, Molecular and Cellular Biology — *8/2016 – 12/2025*

I discovered a novel regulatory mechanism in bacterial cell division (PLoS One, 2025; co-corresponding author).

- **Novel IDR contact site discovery:** Developed a computational framework for predicting interactions between folded domains and intrinsically disordered regions (IDRs) using ensembles of AlphaFold-Multimer models to statistically identify contact residues. Validated my *in silico* predictions experimentally via ITC and quantitative 2D NMR.
- **High-throughput cell morphometric measurements:** Designed and performed live-cell (wide-field, TIRF, confocal) imaging experiments — from microscopy to quantitative image analysis pipelines. Developed computer vision and computational geometry algorithms to quantify cell shape and subcellular dynamics; hand-built and optimized core algorithms in vectorized NumPy; profiled to eliminate bottlenecks; parallelized processing on a SLURM cluster.
- **Proximity-labeling candidate stratification:** Devised a strategy for ranking candidate lipid-droplet-surface proteins from time-resolved proximity-labeling mass-spec, using enrichment kinetics to distinguish recruitment to the droplet surface from static background; 17 of the 20 top-ranked candidates were experimentally validated *in vivo*.

Broad Institute of MIT and Harvard — *Cambridge, MA*
Computational Biologist, Rinn Lab — *2/2013 – 7/2016*

I was a core computational analyst on 11 publications.

- **Functional genomics:** Developed custom integrative multi-omics analyses (RNA-seq, ChIP-seq, RIP-seq, ATAC-seq) to extract molecular mechanisms from perturbations and time courses of stem cell and developmental models; iterated with bench collaborators on experimental design and interpretation.
- **Quantitative biology:** Designed statistical metrics to measure cell-biological features, e.g., mRNA splicing efficiency, export efficiency, and tissue specificity.
- **Computational infrastructure:** Built the lab's workhorse lncRNA discovery pipeline.

Broad Institute of MIT and Harvard — *Cambridge, MA*
Software Engineer, The Cancer Genome Atlas — *4/2012 – 2/2013*

Worked on the GDAC Firehose, a production genomics platform processing ∼14,000 samples across 25 cancer types.

- Identified and eliminated bottlenecks, refactoring or rewriting omics analyses to run at scale.
- Traced one production failure to the Python standard library, and contributed upstream fixes to CPython 3.4 — enabling ZIP64 archives in shutil (#17189) and making them the default in zipfile (#17201).

Earlier Engineering Experience

Space Sciences Laboratory — *Digital Design Engineer*, Townes Lab — *Berkeley, CA · 9/2009 – 8/2011*
Center for Astronomy Signal Processing & Electronics Research — *Digital Design Engineer* — *Berkeley, CA · 4/2008 – 8/2011*
MusicianLink — *Software Engineer* — *San Jose, CA · 6/2007 – 3/2008*
UC Berkeley Clustered Computing — *Unix Systems Administrator* — *Berkeley, CA · 1/2006 – 8/2006*
Massachusetts Institute of Technology — *Research Assistant*, Chuang Lab — *Cambridge, MA · 6/2005 – 8/2005*
University of California, Berkeley — *Research Assistant*, Holzapfel Lab — *Berkeley, CA · 6/2004 – 5/2005*
Berkeley Student Cooperative — *Unix Network Administrator* — *Berkeley, CA · 8/2003 – 12/2006*

Education

Harvard University — PhD, Biochemistry — *Cambridge, MA · 5/2025*
University of California, Berkeley — BS, Engineering Physics; *minors in CS and Mathematics* — *Berkeley, CA · 5/2007*

Recent Projects

bacterial phospho-site predictor (2026) — Trained an MLP on frozen ESM-2 embeddings to predict phospho-sites in bacteria, sweeping hyperparameters to find a minimal architecture. Discovered that a published ten-feature ensemble's benchmark was inflated by train-on-test contamination; showed that my simpler model matched its performance when both were run on a properly split version of their dataset.

bio knowledge graph (2026) — Built a knowledge graph over seven decades of bacterial cell-division literature (∼8k papers), ingesting PubMed abstracts, PMC full text, and scanned pre-digital PDFs via local OCR (MinerU). Tags biomolecules with NER and links them through published vocabularies and ontologies. Co-mention edges point back to their source sentences. Disambiguates authors and institutions. A citation-lineage layer recovers a gene's literature from before its name stabilized.

discourse-analysis (2026) — Built a search engine over a Twitter corpus (∼500k posts) unifying full-text, fuzzy, and semantic search. Runs image OCR (Tesseract), machine translation (NLLB-200), and embedding (Qwen) on-device at zero cloud cost. Retrieves posts and reconstructs their context (threads, replies, quotes). Exposes an agentic tool interface (MCP) tailored to how an LLM explores a corpus.

gpu-xengine (2026) — Ported a matrix multiplication engine for real-time signal cross-correlation from FPGA to CUDA. Optimized memory access patterns (coalescing, pinning) for a 1.5× throughput gain.

Earlier Projects

ISI digital backend (2011) — Real-time FX correlator for the Infrared Spatial Interferometer at Mount Wilson Observatory.

- Designed a 138 Gbps signal processing system across 3 synchronized Xilinx Virtex-5 FPGAs, channelizing 2.88 GHz of bandwidth into 64 spectral channels with real-time cross-correlation.
- Achieved a 1.8× clock speedup (200 → 360 MHz) over all prior CASPER designs by mapping core FFT arithmetic onto DSP48Es and developing a BRAM/DSP floorplanning methodology.
- Contributed 50+ optimized DSP commits upstream to CASPER's open-source FPGA library; documented the floorplanning methodology in CASPER Memo #42.
- Built an end-to-end data acquisition pipeline, from FPGA fabric to PowerPC firmware to Python control software and networked storage.
- Instrument resolved individual molecular signatures in Betelgeuse's atmosphere; results published in Wishnow et al. 2010 (SPIE); code and documentation available on GitHub.

CASPER FPGA toolkit (2009) — Contributed to CASPER, an open-source FPGA toolkit for radio astronomy signal processing.

- Refactored and extended Simulink blocks and mask scripts in the CASPER FPGA library.
- Developed a 10 Gbps UDP receiver with a synchronized ring buffer; the core architecture was adopted into PySPEAD, the data transport layer for the Square Kilometre Array.
- Built enclosure and RF signal routing for a spectrometer deployed at Arecibo Observatory.

Quantum ion-trap pulse sequencer (2005) — Built an instrument to programmatically shuttle charged microspheres around a planar Paul trap (Phys. Rev. A, 2006).

- Designed a compact instruction set for precisely timed voltage sequences; wrote a "pulse script" interpreter in ∼1,000 lines of Scenix microcontroller assembly, and a Python client to drive it.
- Built the hardware: salvaged a NIM enclosure, mounted and wired the microcontroller and buffer electronics, designed a front panel in CAD, and cut it from aluminum stock on a waterjet.
- First planar-trap demonstration of the three movement primitives required for a scalable ion-trap quantum computer.

Helium dilution fridge control system (2004) — Wrote a C++ control system for a helium dilution fridge cooling a millimeter-wave telescope detector to 250 mK.

- Rewrote the original LabVIEW system from scratch in C++ for robustness and remote deployment.
- Interfaced with ADC and GPIO boards via Comedi device drivers on Linux.
- The system sequenced the multi-stage cooldown, then held temperature through closed-loop feedback.

Publications

W.J. Mallard†, V.V. Pham†. "FtsZ phosphorylation modulates tail-core binding to tune cell division in *Bacillus subtilis*." *PLOS ONE* 20(12), e0337820 (2025).

W.J. Mallard. "FtsZ phosphorylation modulates tail-core binding to tune cell division in *Bacillus subtilis*." *PhD thesis, Harvard University*, ProQuest 32042133 (2025).

J. Erber, C. Stecher, V. Plajer, N. Braun, **W. Mallard**, L.A. Goff, I. Barozzi, T. Mohr, J.L. Rinn, R.A. Flavell, D. Herndler-Brandstetter. "*Rroid2* regulates effector-to-memory CD8+ T cell differentiation during infection *in vivo*." *PNAS* 122(48), e2503450122 (2025).

A.K.K. Teo, L. Nguyen, M.K. Gupta, H.H. Lau, L.S.W. Loo, N. Jackson, C.S. Lim, **W. Mallard**, M.A. Gritsenko, J.L. Rinn, R.D. Smith, W.J. Qian, R.N. Kulkarni. "Defective insulin receptor signaling in hPSCs skews pluripotency and negatively perturbs neural differentiation." *Journal of Biological Chemistry* 296, 100495 (2021).

A. Choudhuri, E. Trompouki, B.J. Abraham, L.M. Colli, K.H. Kock, **W. Mallard**, M.L. Yang, D.S. Vinjamur, A. Ghamari, A. Sporrij, K. Hoi, B. Hummel, S. Boatman, V. Chan, S. Tseng, S.K. Nandakumar, S. Yang, A. Lichtig, M. Superdock, S.N. Grimes, T.V. Bowman, Y. Zhou, S. Takahashi, R. Joehanes, A.B. Cantor, D.E. Bauer, S.K. Ganesh, J. Rinn, P.S. Albert, M.L. Bulyk, S.J. Chanock, R.A. Young, L.I. Zon. "Common variants in signaling transcription-factor-binding sites drive phenotypic variability in red blood cell traits." *Nature Genetics* 52(12), 1333–1345 (2020).

J.P. Lewandowski, J.C. Lee, T. Hwang, H. Sunwoo, J.M. Goldstein, A.F. Groff, N.P. Chang, **W. Mallard**, A. Williams, J. Henao-Meija, R.A. Flavell, J.T. Lee, C. Gerhardinger, A.J. Wagers, J.L. Rinn. "The *Firre* locus produces a trans-acting RNA molecule that functions in hematopoiesis." *Nature Communications* 10(1), 5137 (2019).

M. Melé, K. Mattioli, **W. Mallard**, D.M. Shechner, C. Gerhardinger, J.L. Rinn. "Chromatin environment, transcriptional regulation, and splicing distinguish lincRNAs and mRNAs." *Genome Research* 27(1), 27–37 (2017).

P. Kaewsapsak, D.M. Shechner, **W. Mallard**, J.L. Rinn, A.Y. Ting. "Live-cell mapping of organelle-associated RNAs via proximity biotinylation combined with protein-RNA crosslinking." *eLife* 6, e29224 (2017).

L.M.R. Ferreira, T.B. Meissner, T.S. Mikkelsen, **W. Mallard**, C.W. O'Donnell, T. Tilburgs, H.A.B. Gomes, R. Camahort, R.I. Sherwood, D.K. Gifford, J.L. Rinn, C.A. Cowan, J.L. Strominger. "A distant trophoblast-specific enhancer controls HLA-G expression at the maternal-fetal interface." *PNAS* 113(19), 5364–5369 (2016).

J. Choi, S. Lee, **W. Mallard**, K. Clement, G.M. Tagliazucchi, H. Lim, I.Y. Choi, F. Ferrari, A.M. Tsankov, R. Pop, G. Lee, J.L. Rinn, A. Meissner, P.J. Park, K. Hochedlinger. "A comparison of genetically matched cell lines reveals the equivalence of human iPSCs and ESCs." *Nature Biotechnology* 33(11), 1173–1181 (2015).

M.J. Ziller, R. Edri, Y. Yaffe, J. Donaghey, R. Pop, **W. Mallard**, R. Issner, C.A. Gifford, A. Goren, J. Xing, H. Gu, D. Cacchiarelli, A.M. Tsankov, C. Epstein, J.L. Rinn, T.S. Mikkelsen, O. Kohlbacher, A. Gnirke, B.E. Bernstein, Y. Elkabetz, A. Meissner. "Dissecting neural differentiation regulatory networks through epigenetic footprinting." *Nature* 518(7539), 355–359 (2015).

J. Liao, R. Karnik, H. Gu, M.J. Ziller, K. Clement, A.M. Tsankov, V. Akopian, C.A. Gifford, J. Donaghey, C. Galonska, R. Pop, D. Reyon, S.Q. Tsai, **W. Mallard**, J.K. Joung, J.L. Rinn, A. Gnirke, A. Meissner. "Targeted disruption of DNMT1, DNMT3A and DNMT3B in human embryonic stem cells." *Nature Genetics* 47(5), 469–478 (2015).

The Cancer Genome Atlas Research Network. "Comprehensive molecular profiling of lung adenocarcinoma." *Nature* 511(7511), 543–550 (2014).

M. Sauvageau, L.A. Goff, S. Lodato, B. Bonev, A.F. Groff, C. Gerhardinger, D.B. Sanchez-Gomez, E. Hacisuleyman, E. Li, M. Spence, S.C. Liapis, **W. Mallard**, M. Morse, M.R. Swerdel, M.F. D'Ecclessis, J.C. Moore, V. Lai, G. Gong, G.D. Yancopoulos, D. Frendewey, M. Kellis, R.P. Hart, D.M. Valenzuela, P. Arlotta, J.L. Rinn. "Multiple knockout mouse models reveal lincRNAs are required for life and brain development." *eLife* 2, e01749 (2013).

The Cancer Genome Atlas Research Network. "The somatic genomic landscape of glioblastoma." *Cell* 155(2), 462–477 (2013).

The Cancer Genome Atlas Research Network. "Comprehensive molecular characterization of clear cell renal cell carcinoma." *Nature* 499(7456), 43–49 (2013).

The Cancer Genome Atlas Research Network. "Integrated genomic characterization of endometrial carcinoma." *Nature* 497(7447), 67–73 (2013).

A. Siemion, H. Chen, J. Cobb, J. Cordes, T. Filiba, A. Fries, A. Howard, J. von Korff, E. Korpela, M. Lebofsky, **W. Mallard**, P. McMahon, A. Parsons, L. Spitler, M. Wagner, D. Werthimer. "Current and nascent SETI instruments in the radio and optical." *Communication with Extraterrestrial Intelligence (CETI)*, pp. 19–36 (2011).

E.H. Wishnow, **W. Mallard**, V. Ravi, S. Lockwood, W. Fitelson, D. Werthimer, C.H. Townes. "Mid-infrared interferometry with high spectral resolution." *Optical and Infrared Interferometry II (Proc. SPIE)* 7734, 773409 (2010).

C.E. Pearson, D.R. Leibrandt, W.S. Bakr, **W.J. Mallard**, K.R. Brown, I.L. Chuang. "Experimental investigation of planar ion traps." *Physical Review A* 73(3), 032307 (2006).

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Conference Proceedings & Abstracts

J. Erber, C. Stecher, V. Plajer, N. Braun, **W. Mallard**, L. Goff, J. Zhao, et al. "Long noncoding RNA-124 regulates CD8+ T cell response to infection." *European Journal of Immunology* 54, 263 (2024).

A. Choudhuri, E. Trompouki, B.J. Abraham, L. Colli, **W. Mallard**, M.L. Yang, D. Vinjamur, A. Ghamari, S. Nandakumar, K. Hoi, B. Hummel, S. Boatman, V. Chan, T.V. Bowman, S. Yang, Y. Zhou, S. Takahashi, A.B. Cantor, V.G. Sankaran, S. Ganesh, D.E. Bauer, J. Rinn, S.J. Chanock, R.A. Young, L.I. Zon. "Transcriptional signaling centers govern human erythropoiesis and harbor genetic variations of red blood cell traits." *Blood* 132(Suppl 1), 1277 (2018).

A. Choudhuri, E. Trompouki, B.J. Abraham, **W. Mallard**, M.L. Yang, A. Ghamari, K. Hoi, B. Hummel, S. Boatman, V. Chan, T.V. Bowman, S. Yang, Y. Zhou, S. Takahashi, A.B. Cantor, S. Ganesh, J. Rinn, R.A. Young, L.I. Zon. "Distinct signaling centers define stages of human erythropoiesis and harbor common variations of red blood cell traits." *Blood* 130(Suppl 1), 773 (2017).

L. Ferreira, T. Meissner, T. Mikkelsen, C. O'Donnell, R. Sherwood, **W. Mallard**, J. Rinn, C. Cowan, J. Strominger. "Long-range chromatin interactions control trophoblast-restricted HLA-G expression during pregnancy." *The Journal of Immunology* 194(1 Suppl), 60.7 (2015).

A. Choudhuri, B. Abraham, **W. Mallard**, B. Hummel, S. Boatman, T. Bowman, A. DiBiase, J. Rinn, R. Young, L. Zon, E. Trompouki. "Lineage regulators and signaling transcription factors during erythropoiesis." *Experimental Hematology* 42(8), S63 (2014).

B. Lehnertz, G. Boucher, **W. Mallard**, P. Gendron, J. Hebert, S. Lemieux, J. Rinn, G. Sauvageau. "Long non-coding RNA expression in normal and leukemic human blood samples." *Experimental Hematology* 42(8), S45 (2014).

A.P.V. Siemion, D. Werthimer, D. Anderson, H. Chen, J. Cobb, J. Cordes, T. Filiba, G. Foster, S. Gowda, E. Korpela, M. Lebofsky, A. Little, **W. Mallard**, L. Spitler, M. Wagner. "Developments in the radio search for extraterrestrial intelligence." *URSI General Assembly and Scientific Symposium*, pp. 1–4 (2011).

D.A. Roshi, M. Bloss, P. Brandt, S. Bussa, H. Chen, P. Demorest, G. Desvignes, T. Filiba, R.J. Fisher, J. Ford, D. Frayer, R. Garwood, S. Gowda, G. Jones, **W. Mallard**, J. Masters, R. McCullough, G. Molera, K. O'Neil, J. Ray, S. Scott, A. Shelton, A. Siemion, M. Wagner, G. Watts, D. Werthimer, M. Whitehead. "Advanced multi-beam spectrometer for the Green Bank Telescope." *URSI General Assembly and Scientific Symposium* (2011).

T. Filiba, H. Chen, S. Gowda, **W. Mallard**, J. Manley, P. McMahon, A. Siemion, L. Spitler, M. Wagner, D. Werthimer. "Wideband FPGA spectrometers and correlators." *USNC/URSI National Radio Science Meeting*, p. 32 (2009).

Teaching Experience

Harvard University — *Teaching Fellow*, MCB 112: Biological Data Analysis — *Cambridge, MA · Fall 2021*

Harvard University — *Lecturer*, MCB Matlab Bootcamp — *Cambridge, MA · Spring 2019*

Harvard University — *Teaching Fellow*, MCB 112: Biological Data Analysis — *Cambridge, MA · Fall 2017*

Harvard University — *Lecturer*, MCB Matlab Bootcamp — *Cambridge, MA · Fall 2017*

Harvard University — *Guest Lecturer*, SCRB 157: The Noncoding Genome — *Cambridge, MA · Spring 2014*

UC Berkeley — *Lab Teaching Assistant*, CS 61B: Data Structures — *Berkeley, CA · Fall 2007*